

VILAASA ESTATES PLATFORM

MASTER ENGINEERING HANDOVER & DEVELOPER ROADMAP

SYSTEM OVERVIEW

1. Executive Overview & Architecture

Vilaasa Estates is an enterprise-grade luxury real estate and fractional franchise platform designed for high-net-worth individuals, institutional capital desks, channel partners, and internal asset directors. The platform provides a seamless bridge between Dubai and Indian trophy properties with four specialized isolated application portals.

Core Portals & Access Control

- **Public Luxury Showcase (Port 8080):** Client-facing responsive portal featuring immersive property dossiers, wealth return projection calculator, site inspection calendar, franchise investments, and live concierge.
- **Super Admin Command Center (/admin):** Executive management portal with full CRUD for property portfolios, Cloudinary asset uploads, brochure management, inquiry pipeline stages, site visit itineraries, and partner directory.
- **Channel Partner Brokerage Portal (/partner):** Specialized workspace for accredited brokers to browse portfolio inventory, generate VIP referral links, register investor leads, and dispatch client site inspections.
- **The Vault Investor Portal (/vault):** Private custodial portal for registered property owners with 6 live marked-to-market KPI metrics, portfolio holdings, rental yield tracking, and multi-currency formatter.

Technology Stack & Infrastructure

- **Backend:** Node.js, Express.js, TypeScript, PostgreSQL, Prisma ORM, JWT, Bcrypt, Multer, Nodemailer.
- **Frontend:** React 18, Vite, TypeScript, TailwindCSS, Framer Motion, TanStack Query, Lucide Icons, Shadcn UI.
- **Cloud & Storage:** Cloudinary CDN (High-res media & PDF brochures), Gmail SMTP (Transactional emails).

WHAT WAS ACCOMPLISHED

ENGINEERING MILESTONES & REFACTORING

IMPLEMENTATION DETAILS

2. Comprehensive Work Completed

A. Saleor GraphQL Eradication & REST API Migration

- **Native Express REST API:** Replaced all legacy Saleor GraphQL queries (`api.theeyelevelstudio.com`) with high-performance Express REST endpoints (`/api/v1/properties`).
- **Franchise REST Hooks:** Rewrote `useNewFranchise.ts` to query Express backend with FOCO model transforms and curated fallback data structures.
- **Dead Code Purge:** Deleted legacy Saleor files (`graphql.ts`, `useProperties.ts`) and archived 20 legacy migration scripts with 0 residual dependencies.

B. Channel Partner Portal Implementation

- **Dedicated Workspace:** Built `PartnerLogin`, `PartnerRegister`, `PartnerLayout`, `PartnerDashboard`, `PartnerInventory`, `PartnerSiteVisits`, and `PartnerLeads`.
- **Lifecycle Approval Flow:** New brokers register via public portal, Super Admin reviews and updates status to APPROVED, automatically triggering an approval email with login credentials.
- **VIP Link Engine:** Brokers generate custom tracked client referral links with pre-populated broker attribution.

C. The Vault Private Investor Portal

- **Security & Role Gating:** Introduced `VAULT_CLIENT` role in Prisma schema with strict JWT validation rejecting non-investor roles.
- **Wealth Command Dashboard:** Built 6 KPI metric cards (Properties Owned, Total Invested, Current Valuation, Capital Gains, Monthly Rental Yield, Annualized Yield).
- **Admin Vault Manager:** Added a dedicated Vault tab in `AdminPropertyDetail.tsx` for assigning property units to investor folios with inline valuation updates.

D. Transactional Email & Security Infrastructure

- **Live SMTP Integration:** Configured Gmail SMTP with automated delivery for site visit itineraries, contact inquiries, and partner onboarding.
- **Fixed Form Validations:** Replaced hardcoded fake email fallbacks (`@investor.com`, `@vip-client.com`) with strictly validated client inputs.
- **Portal Token Isolation:** Fixed Axios interceptors to dynamically resolve portal tokens and avoid cross-portal 401 redirect loops.

DEVELOPER & OPERATOR GUIDE

SETUP, TESTING & COMMANDS

DEVELOPER QUICKSTART

3. How to Run & Work on the Platform

1. Environment Configuration

Ensure vilaasa-backend/.env and vilaasa-frontend/.env are configured:

```
DATABASE_URL=postgresql://postgres:Naif2003@localhost:5432/vilaasa
JWT_SECRET=vilaasa_luxury_secret_jwt_key_2026_super_secure
```

2. Database Synchronization & Seeding

Run the following commands in vilaasa-backend/ to apply migrations and seed initial data:

```
npx prisma generate
npm run db:seed
```

3. Starting Development Servers

- **Backend Server:** In vilaasa-backend: npm run dev (Runs on http://localhost:5000)
- **Frontend Server:** In vilaasa-frontend: npm run dev (Runs on http://localhost:8080)

4. Default Portal Credentials

- **Super Admin (/admin/login):** superadmin@vilaasa.com | SuperAdmin@Vilaasa2026
- **Channel Partner (/partner/login):** partner@luxuryestates.com | Partner@Vilaasa2026
- **Vault Investor (/vault/login):** investor@vilaasa.com | investor123

5. Core API Endpoints Reference

- **Auth Endpoints:** POST /api/v1/auth/login, POST /api/v1/auth/register, POST /api/v1/vault/login
- **Properties:** GET /api/v1/properties, GET /api/v1/properties/:slug, POST /api/v1/properties
- **Vault Portfolio:** GET /api/v1/vault/portfolio, GET /api/v1/vault/assets, POST /api/v1/vault/assets
- **Site Visits & Inquiries:** POST /api/v1/site-visits, POST /api/v1/inquiries, GET /api/v1/channel-partners

STRATEGIC ROADMAP

FUTURE CAPABILITIES & ARCHITECTURE GUIDELINES

FUTURE ENHANCEMENTS

4. Strategic Next Steps & Roadmap

Phase 1: Payment & Escrow Integration

- **Booking Deposits:** Integrate Stripe and Razorpay SDKs for token booking deposits (e.g., 15,00,000 / AED 25,000 lock-in deposits).
- **Escrow Milestone Tracking:** Connect investor payments to construction progress stages with automated tranche drawdowns.

Phase 2: Real-Time Concierge & Chat (WebSockets)

- **Live Portfolio Messaging:** Embed real-time Socket.io chat between Vault Investors and designated Senior Relationship Directors.
- **Push Notifications:** Real-time alert dispatch for marked-to-market valuation hikes and quarterly dividend disbursements.

Phase 3: Interactive 3D Virtual Tours

- **Matterport / Three.js:** Embed interactive 360-degree walkthroughs and architectural floorplan 3D viewer in property dossiers.

Phase 4: Advanced Lead Intelligence & Funnels

- **Broker Commission Tracking:** Automated commission ledger calculating 2%-3% brokerage fee upon closed transactions.
- **Audit Log Exports:** CSV / PDF export functionality for site visit schedules, lead pipeline analytics, and investor statements.

DEPLOYMENT CHECKLIST

5. Staging & Deployment Protocol

Before deploying to staging or production environments, execute:

```
git add .
git commit -m "feat: complete vault portal + channel partner + rest migration"
git push origin main
```